第十五讲: 预训练与视觉基础模型

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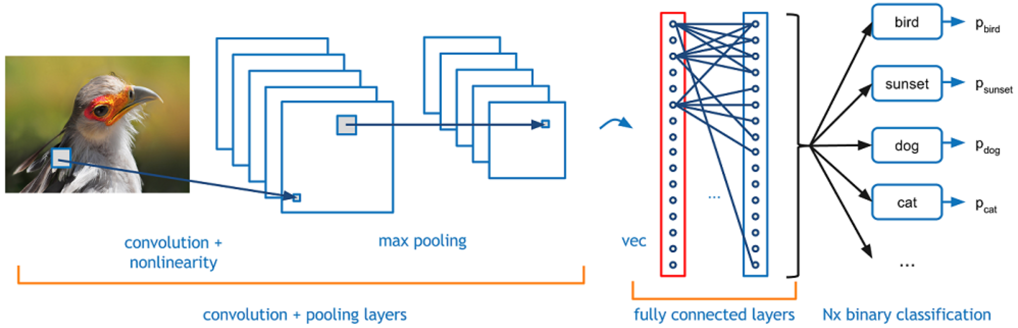
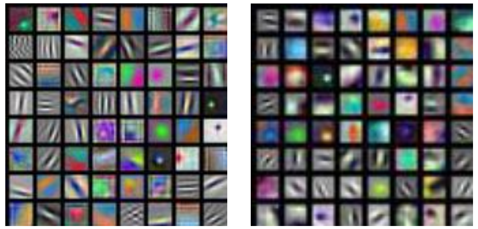
知识回顾

- 卷积神经网络



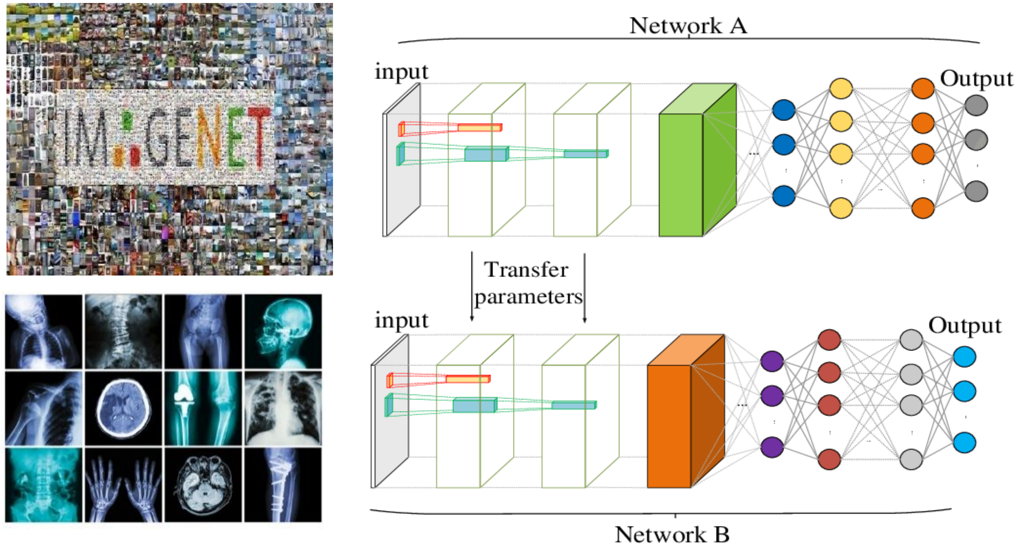


局部特征提取能力
长距离依赖关系表达能力较差

↓

Transformer

- 迁移学习



ImageNet上的预训练

- 需要大量的标注
- 预训练特征偏向于图像分类

↓

自监督预训练

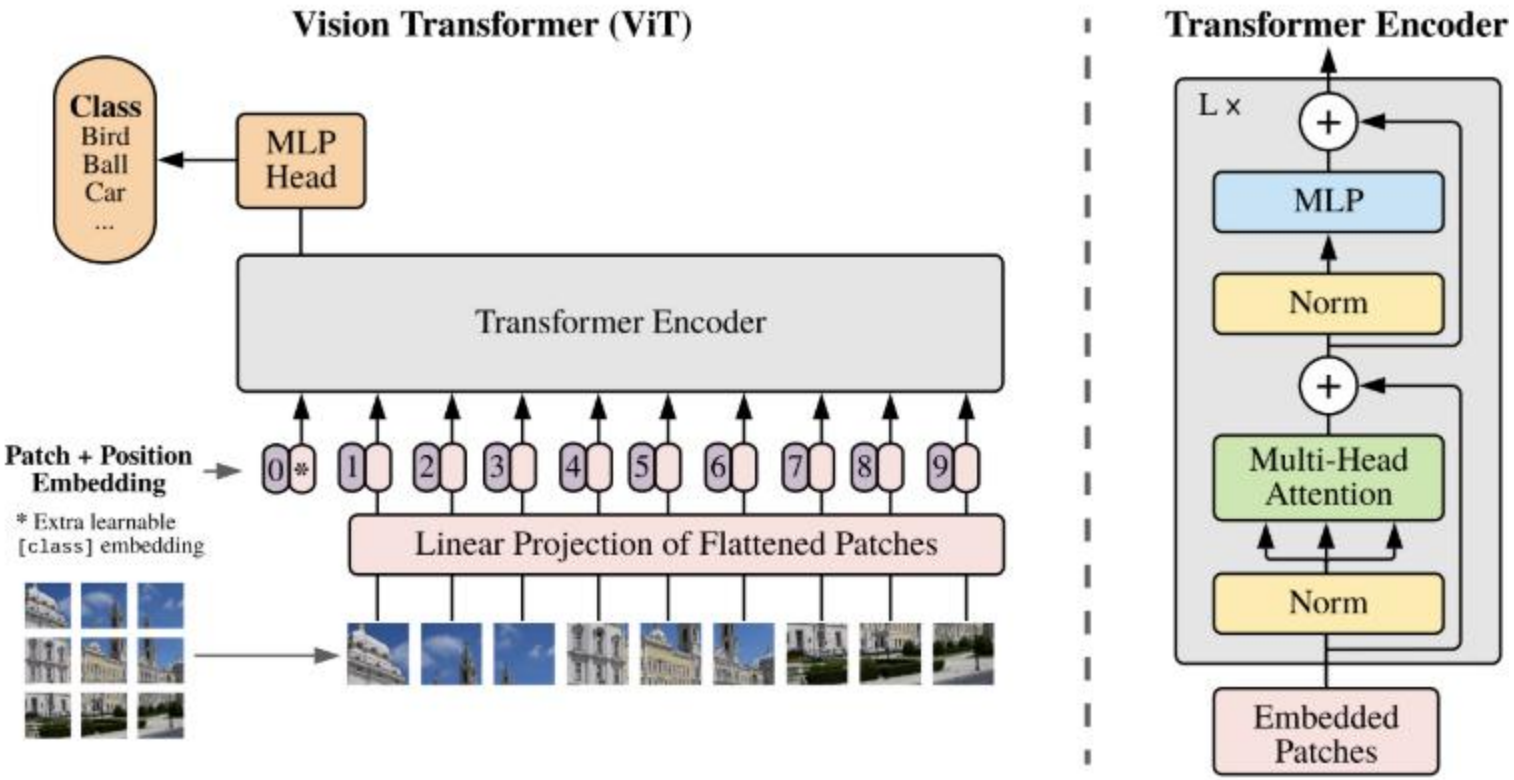
Transformer

- Vision Transformer
  - Instead of using Convolution, use transformer for long-range feature capture

1, 将图像划分为块（patch）

2, 从各个块中提取特征（patch embedding）

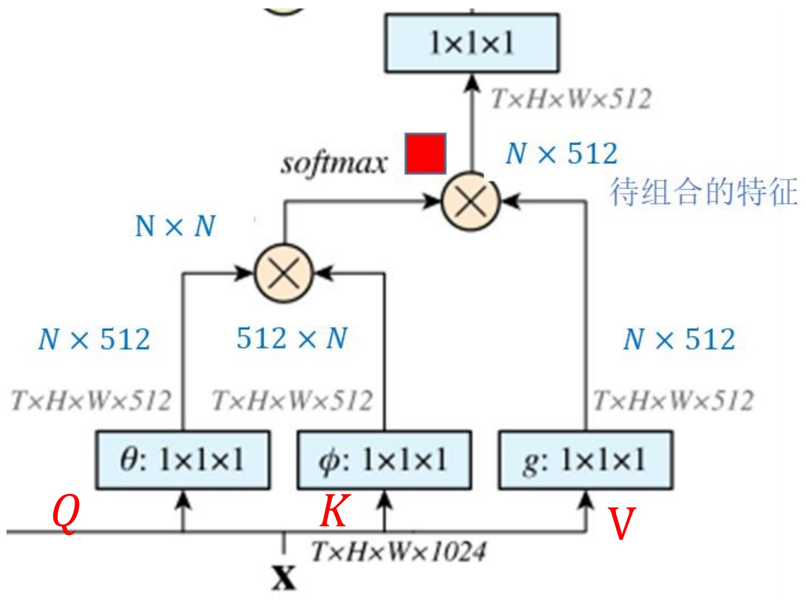
3, 利用多头自注意力机制学习所有图像块的全局依赖关系



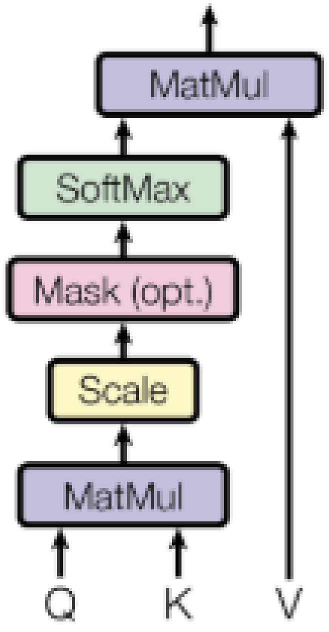
A. Dosovitskiy et al., “An image is worth 16x16 words: Transformers for image recognition at scale,” ICLR, 2021.

Transformer

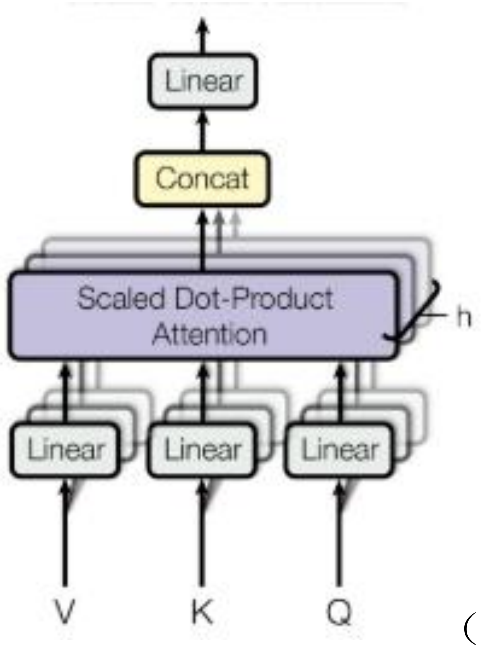
- Multi-Head Self-attention (MHSA) 多头自注意力机制



Non-local中的自注意力



自注意力



多头自注意力

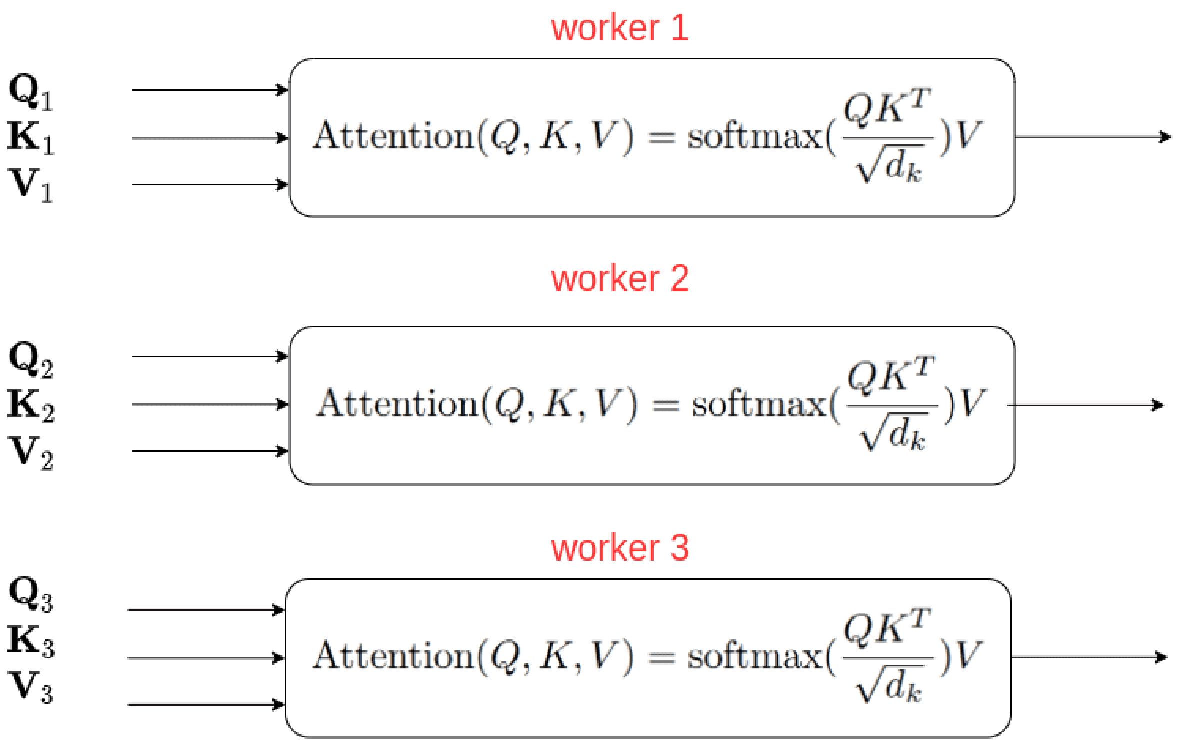
（将通道分组，在各个组上使用自注意力）

A. Dosovitskiy et al., “An image is worth 16x16 words: Transformers for image recognition at scale,” ICLR, 2021.

Transformer

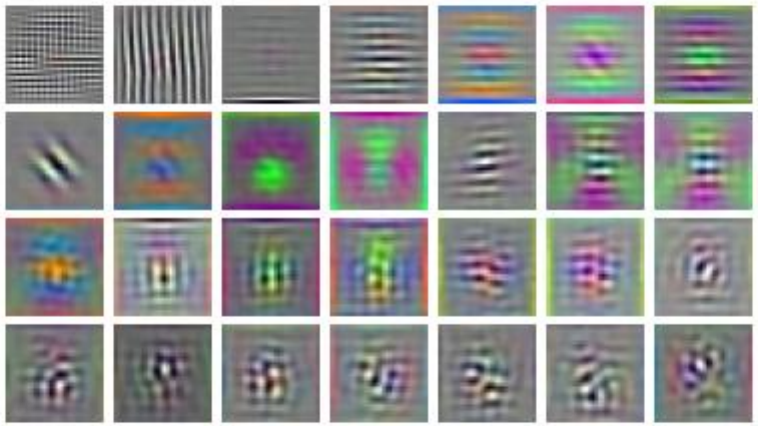
- Multi-Head Self-attention (MHSA) 多头自注意力机制

Each attention head can be implemented in parallel



- 使用多个并行的注意力有助于提升模型的特征表达能力

RGB embedding filters
(first 28 principal components)



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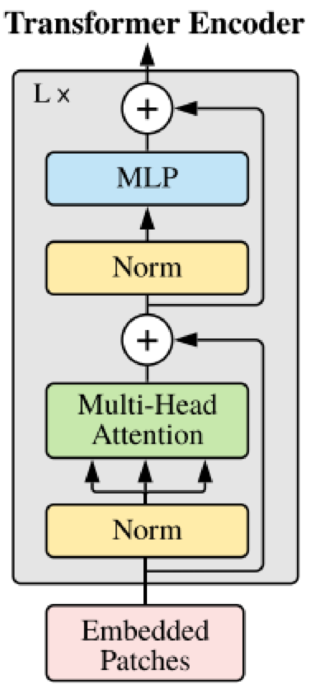
Transformer



- Advantages
  - Long range feature caption
  - Higher performance when trained from large-scale data

- Disadvantage
  - Computationally cost (especially for large image, and small patch size)
  - Large model size (due to use of a lot of MLP layers)
  - Loss of details (patch embedding)

目录

1, Transformer

2，自监督学习

3，视觉基础模型简介

Self-Supervised Learning

- Self-supervised learning methods solve "pretext" tasks that produce **good features** for downstream tasks.
  - Learn with supervised learning objectives, e.g., classification, regression.
  - Labels of these pretext tasks are generated *automatically*

自监督学习通过一个"借口"任务，在不需要人工提供标签的条件下进行特征学习，并且学习到的特征具有较好的通用性，可以迁移到下游任务

- ***Comparison with generative models***

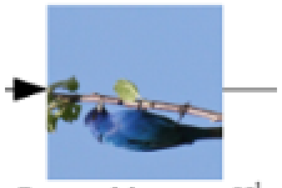
Both aim to learn from data without manual label annotation

Generative learning aims to model data distribution P*data*(*x*), e.g., generating realistic images

Self-Supervised Learning

- 常见的自监督学习方法

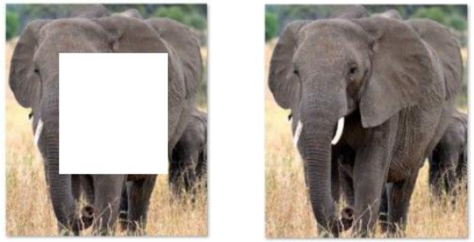
图像分类任务



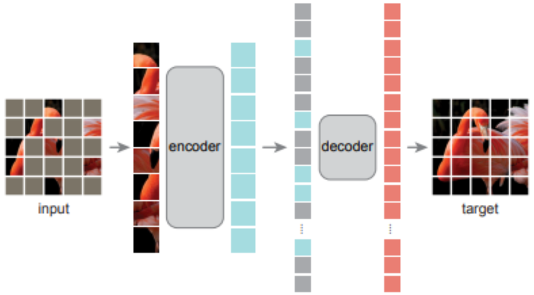
旋转角度分类

图像块顺序分类

图像重建任务

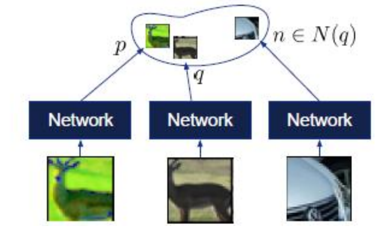


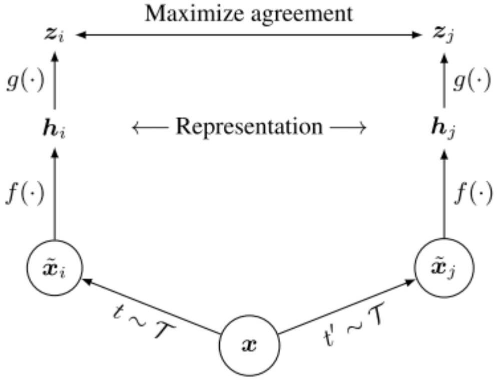
图像复原



掩码重建

对比学习



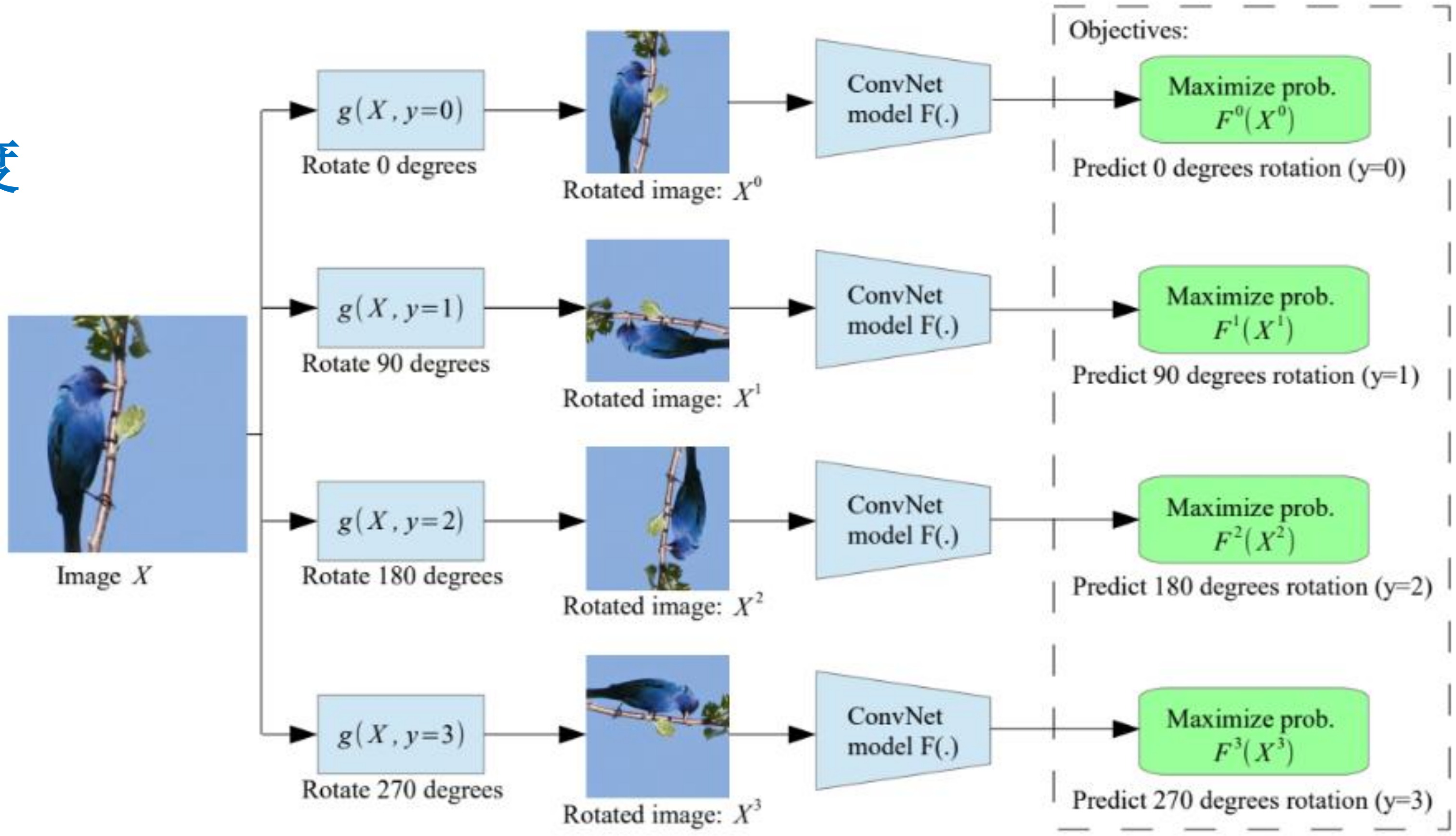


Self-Supervised Learning

- Example: Rotation Prediction

预测图像的旋转角度

正确地识别需要模型理解图像的内容和结构

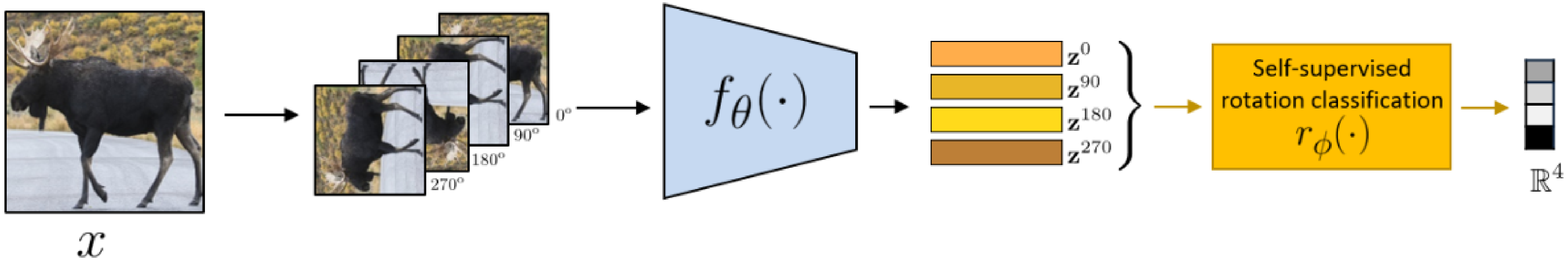


S. Gidaris et al., Unsupervised Representation Learning by Predicting Image Rotations, ICLR 2018

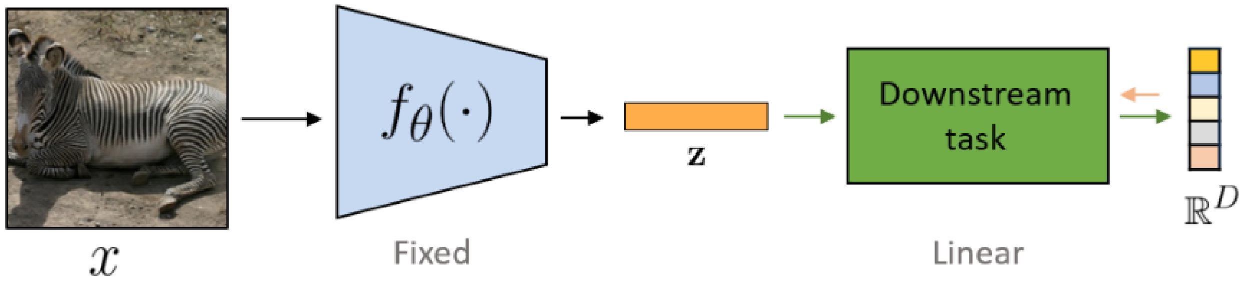
Self-Supervised Learning

- Pipeline

Stage 1: Train network on pretext task (without human labels) 自监督预训练阶段



Stage 2: Train classifier on learned features for new task with fewer labels 下游任务微调阶段

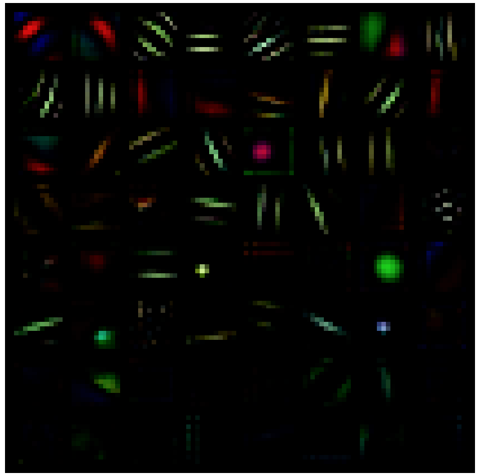


S. Gidaris et al., Unsupervised Representation Learning by Predicting Image Rotations, ICLR 2018

Self-Supervised Learning

- Pretext task: Rotation Prediction

Learned filters in the first layer



Supervised

Self-Supervised

与有监督训练过程学习到的特征类似

Results on CIFAR-10

Model	ConvB1	ConvB2	ConvB3	ConvB4	ConvB5
RotNet with 3 conv. blocks	85.45	88.26	62.09	-	-
RotNet with 4 conv. blocks	85.07	89.06	86.21	61.73	-
RotNet with 5 conv. blocks	85.04	**89.76**	86.82	74.50	50.37

不同层的特征在下游任务中的作用有所差异

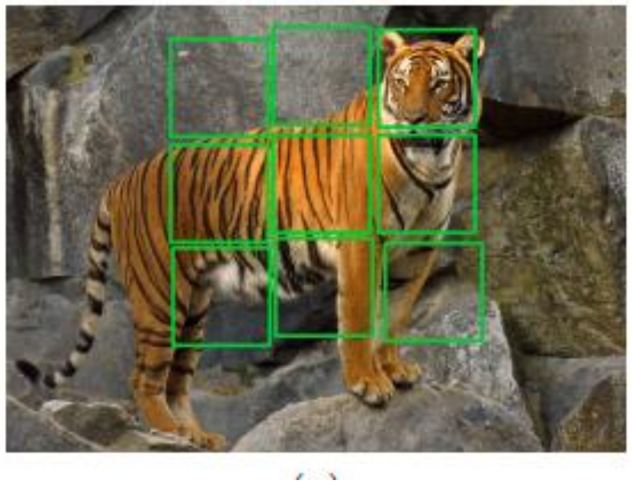
Use the pre-trained model as a feature extractor

Train a non-linear object classifier on top of them

S. Gidaris et al., Unsupervised Representation Learning by Predicting Image Rotations, ICLR 2018

Self-Supervised Learning

- Pretext task: Jigsaw Puzzle



Tiles are extracted

Tile shuffling

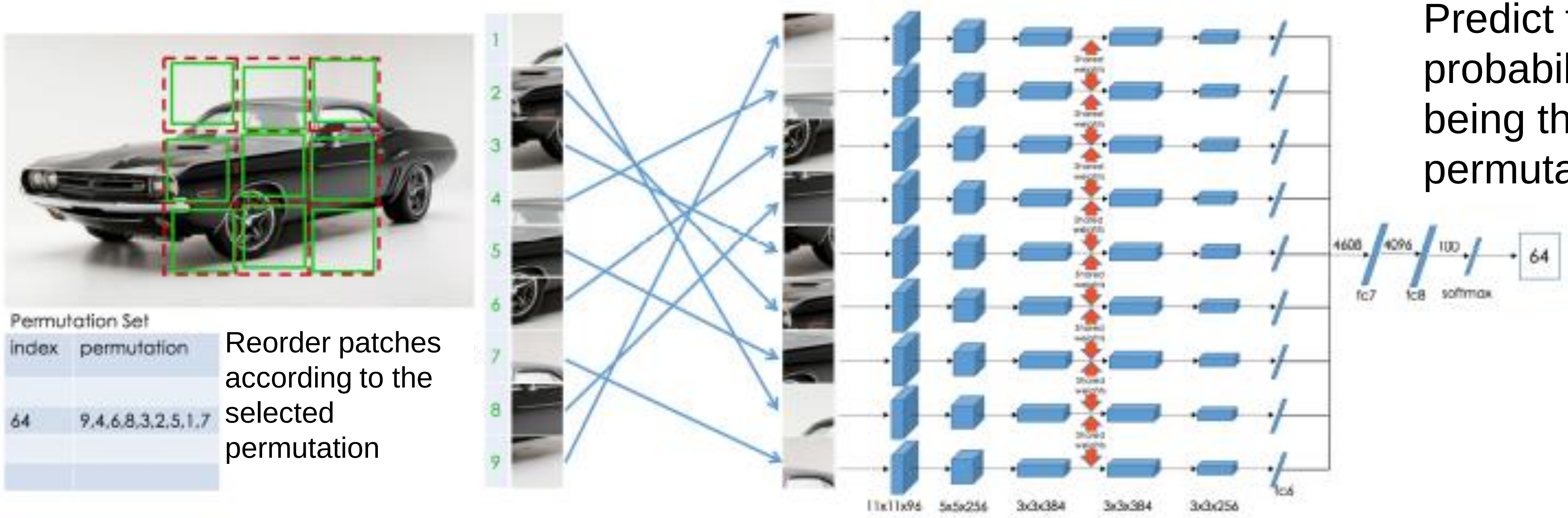
Determining the relative position (challenging)

增强模型对图像的上下文结构的理解

M. Noroozi et al., Unsupervised Learning of Visual Representations by Solving Jigsaw Puzzles, 2017

Self-Supervised Learning

- Pretext task: Jigsaw Puzzle



Reorder patches according to the selected permutation

Predict the probability of being the 64-th permutation

设置64种顺序

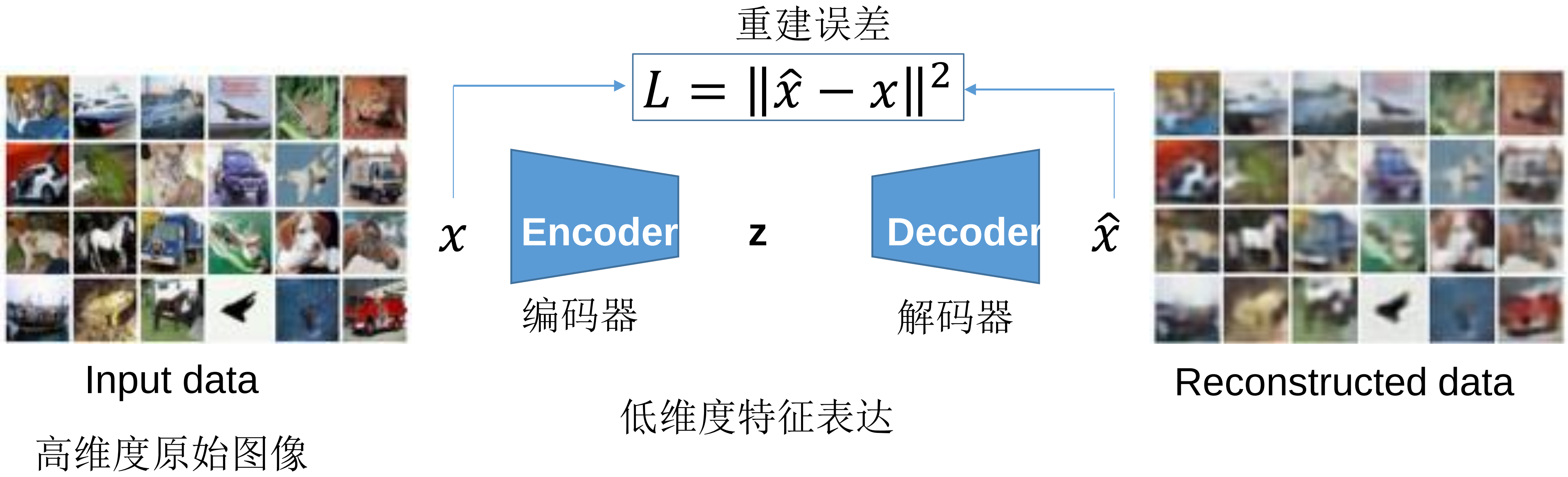
Patch重排

判断是哪一个顺序

M. Noroozi et al., Unsupervised Learning of Visual Representations by Solving Jigsaw Puzzles, 2017

Self-Supervised Learning

- Pretext task: Image reconstruction 图像重建
  - ➢ 自编码器：一种自监督学习方法，可从无标注数据中学习数据的低维表达



Self-Supervised Learning

• Pretext task: Image reconstruction

去噪自编码器

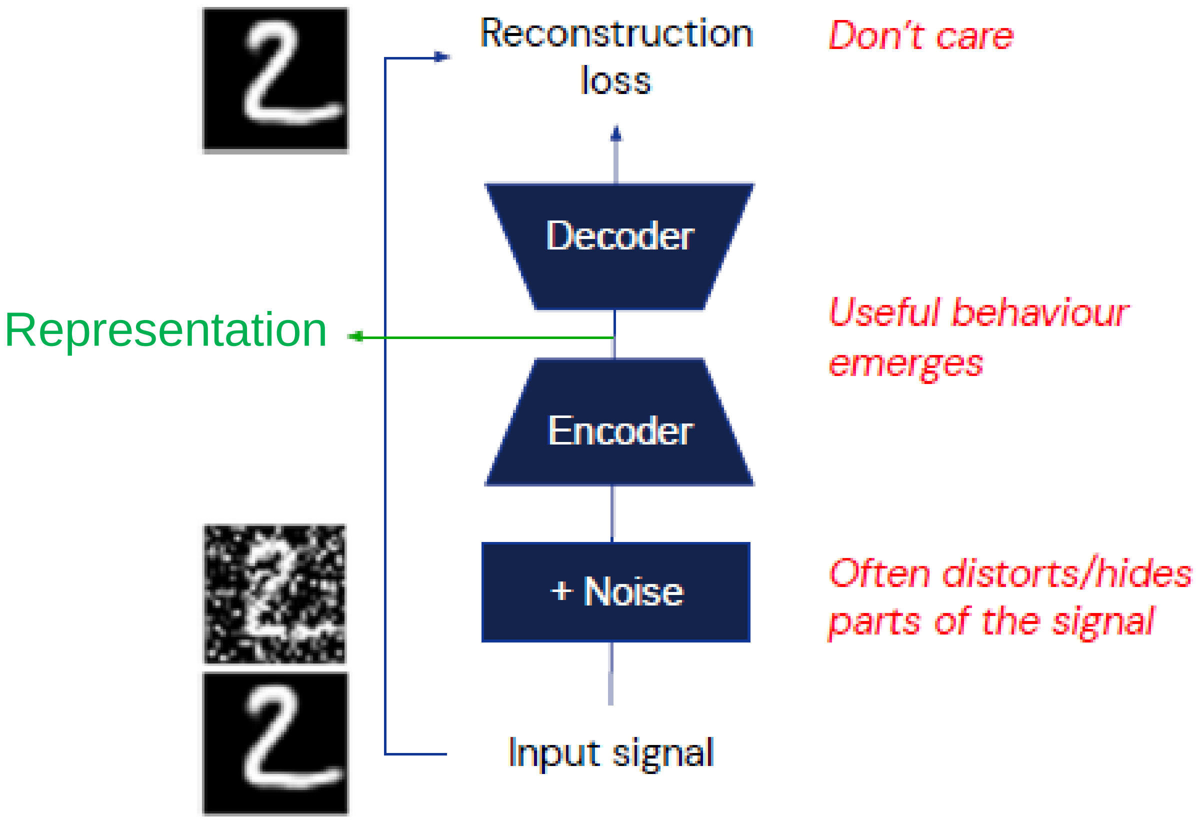
Denoising autoencoders

Pros

- Simple classical method
- Apart from representations, we get a denoiser for free

Cons

- Train-eval gap: training on noisy data
- Too easy, no need for semantics (low level cues are sufficient)



Self-Supervised Learning

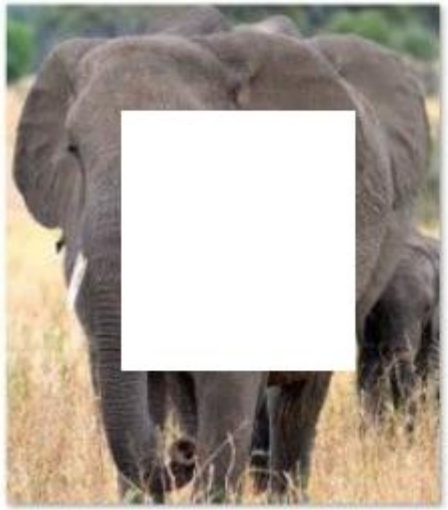
- Pretext task: Image reconstruction

上下文编码器

Context encoders

What goes in the middle?

Much easier if you recognize the objects!



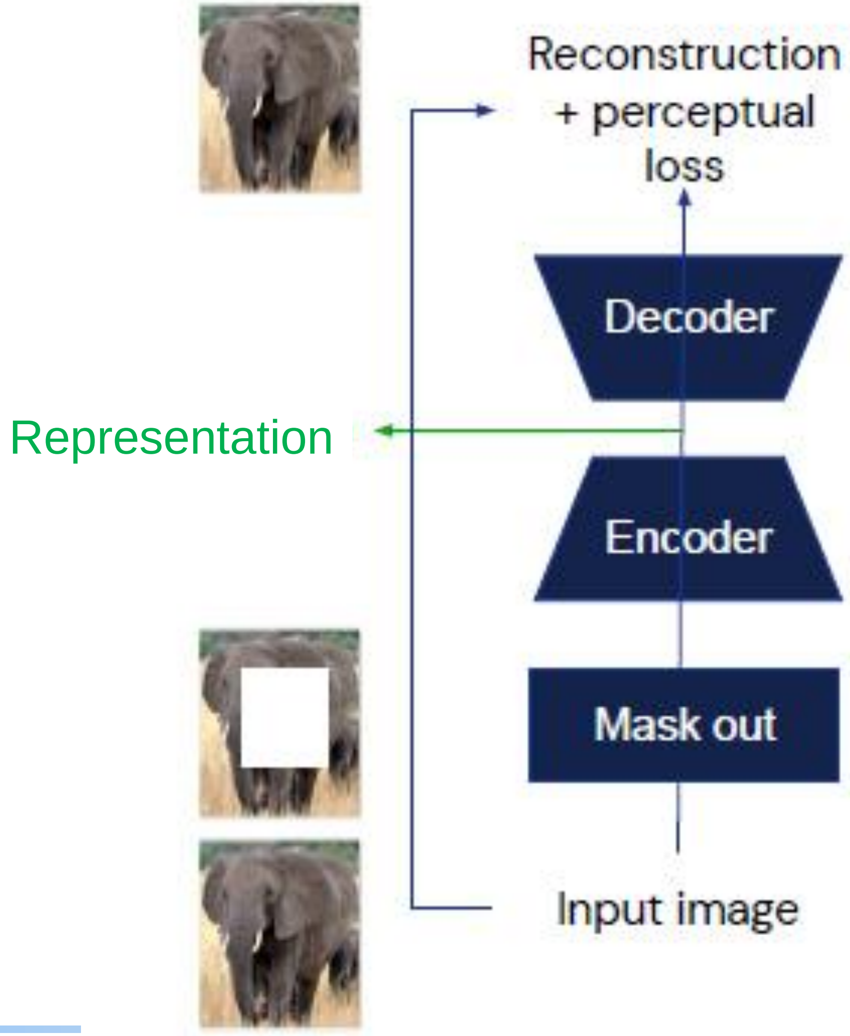
Natural language processing (e.g. word2vec, BERT)

`All [MASK] have tusks. ⇒ All elephants have tusks.`

Self-Supervised Learning

- Pretext task: Image reconstruction

上下文编码器
Context encoders



Pros
- Requires preservation of fine-grained information

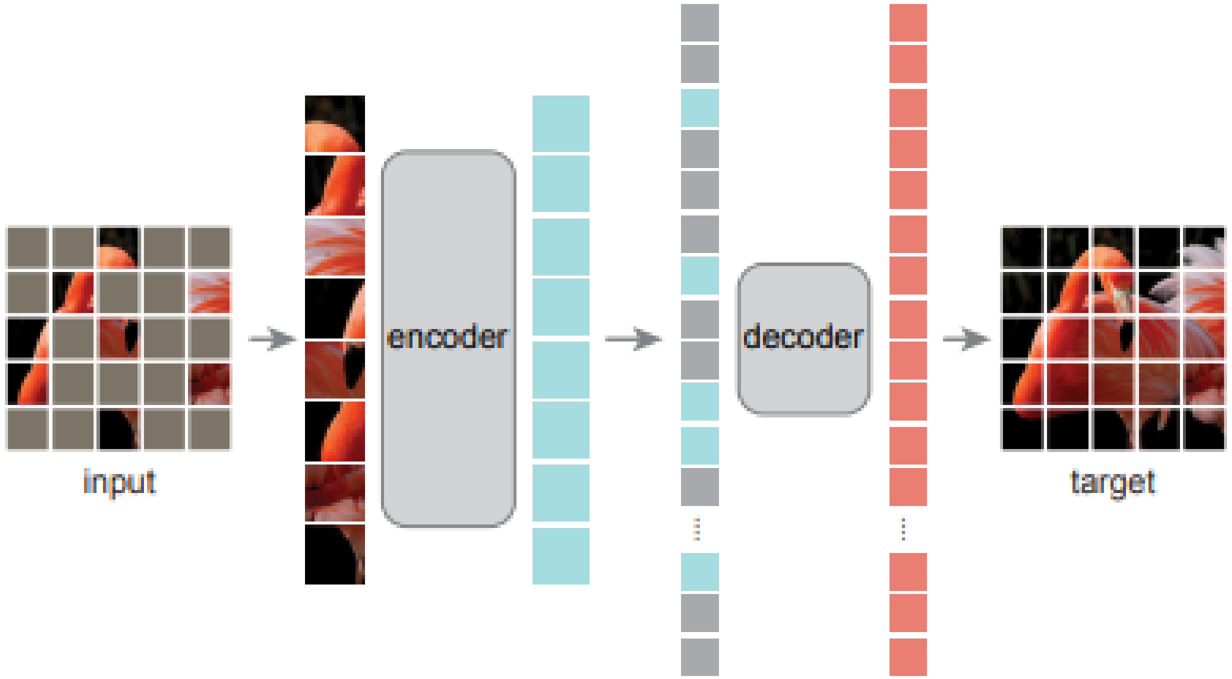
Cons
- Train-eval gap: no masking at eval
- Reconstruction is too hard and ambiguous
- Lots of effort spent on "useless" details: exact colour, good boundary, etc

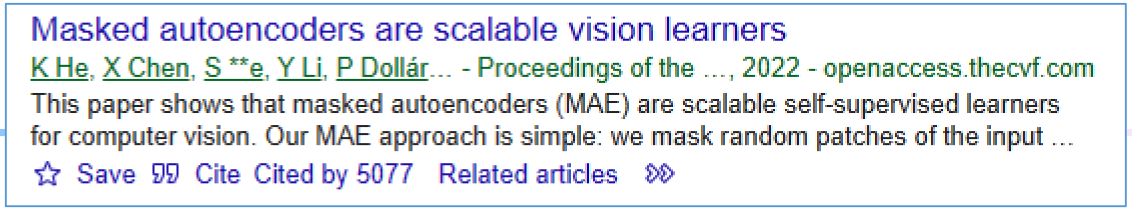
Self-Supervised Learning

- Pretext task: Image reconstruction

Masked AutoEncoders (MAE)

- Proposed for pre-training large transformer models
- Asymmetric encoder-decoder architecture
- High masking ratio (75%) is a meaningful self-supervision task
- Transfer performance in downstream tasks outperforms supervised pretraining




Masked autoencoders are scalable vision learners
K He, X Chen, S **e, Y Li, P Dollár... - Proceedings of the ..., 2022 - openaccess.thecvf.com
This paper shows that masked autoencoders (MAE) are scalable self-supervised learners for computer vision. Our MAE approach is simple: we mask random patches of the input ...
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K. He et al. Masked Autoencoders Are Scalable Vision Learners, 2021

Self-Supervised Learning

- Pretext task: Image reconstruction

Masked AutoEncoders (MAE)

Masked input (80%) and reconstruction

Different masking ratios used for training and testing

K. He et al. Masked Autoencoders Are Scalable Vision Learners, 2021

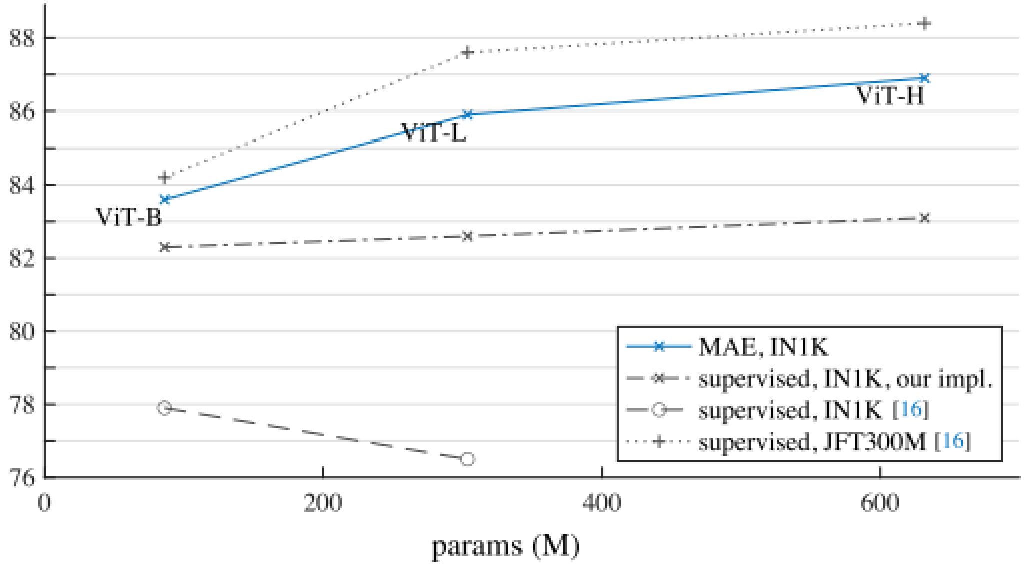
Self-Supervised Learning

- Performance of MAE

Comparisons with previous results on ImageNet- 1K.

method	pre-train data	ViT-B	ViT-L	ViT-H	ViT-H_{448}
scratch, our impl.	-	82.3	82.6	83.1	-
DINO [5]	IN1K	82.8	-	-	-
MoCo v3 [9]	IN1K	83.2	84.1	-	-
BEiT [2]	IN1K+DALLE	83.2	85.2	-	-
MAE	IN1K	83.6	85.9	86.9	**87.8**

MAE pre-training vs. supervised pre-training



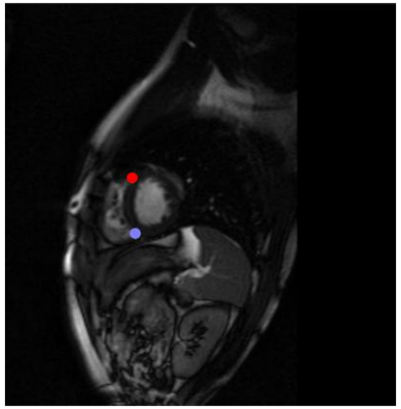
K. He et al. Masked Autoencoders Are Scalable Vision Learners, 2021

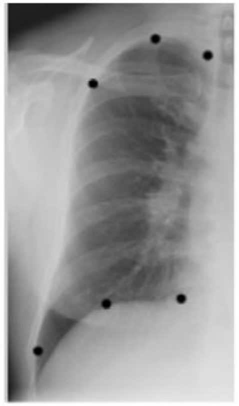
Self-Supervised Learning

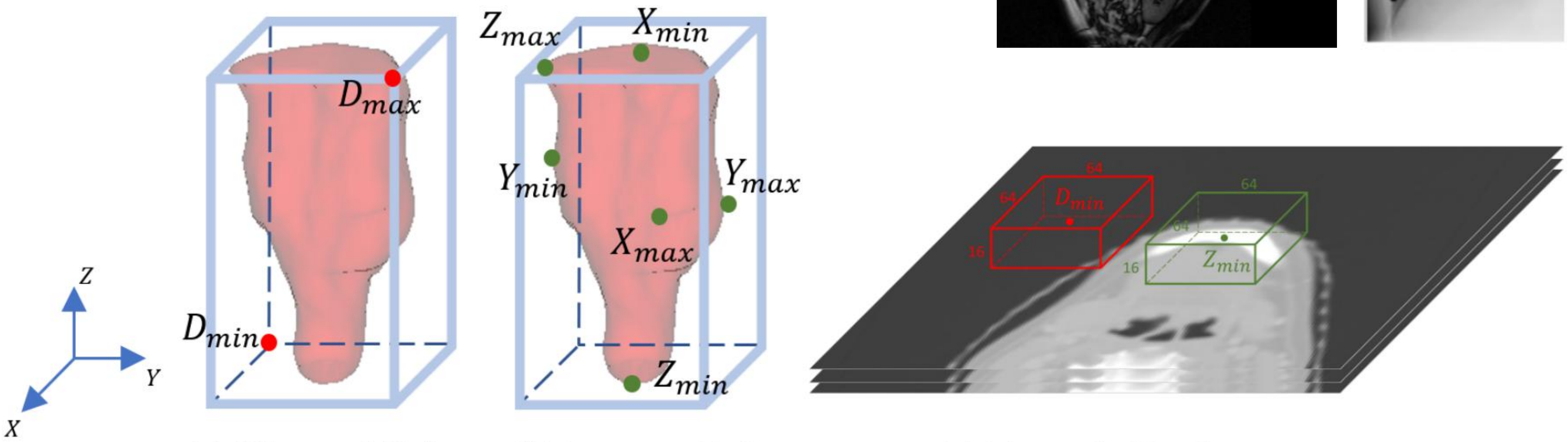
- Pretext task: Relative position prediction 医学图像中的相对位置预测

Background: landmark localization and object detection

Any ideas?







(a) Diagonal Point (b) Extreme Point (c) Example Patches

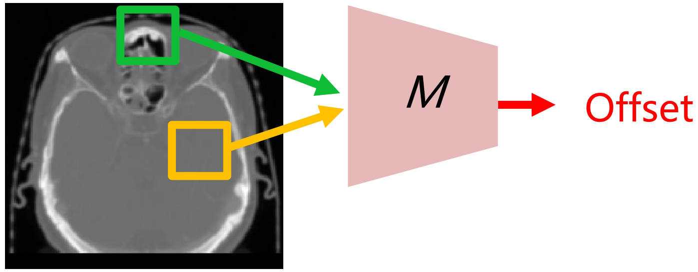
W. Lei et al. Contrastive learning of relative position regression for one-shot object localization in medical images, MICCAI, 2021

Self-Supervised Learning

- Pretext task: Relative position prediction 医学图像中的相对位置预测

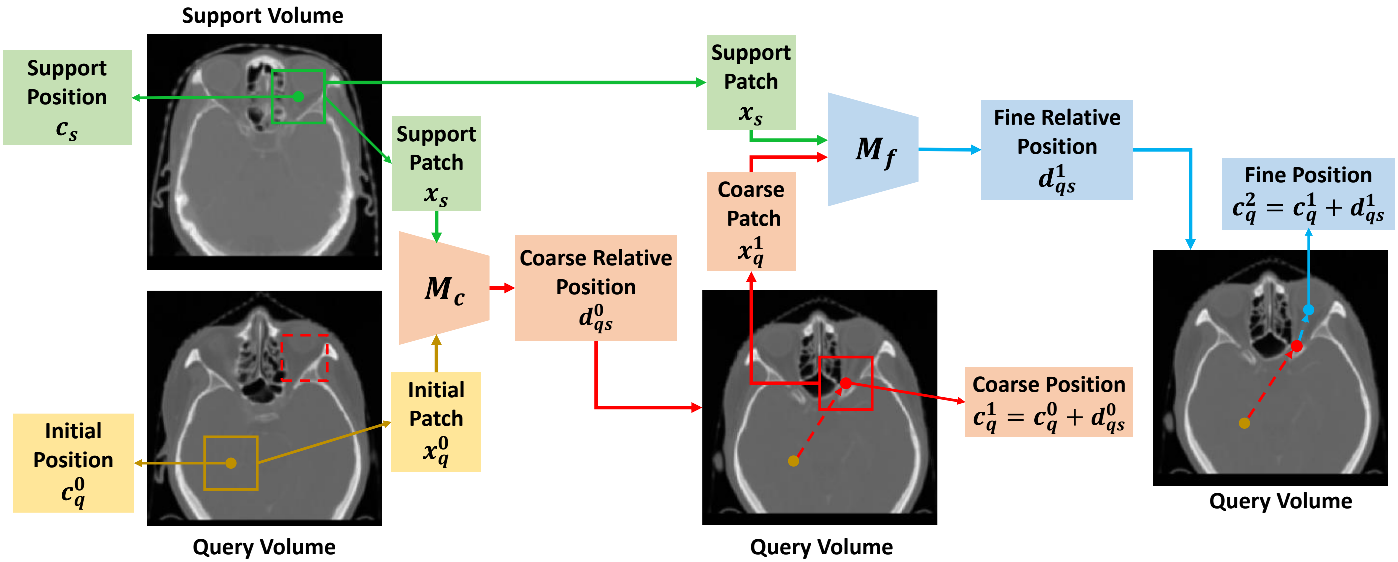
Training stage

- Predicted the offset between patches



Testing stage

- Given a landmark in a reference image, predict the offset from a random position to the target



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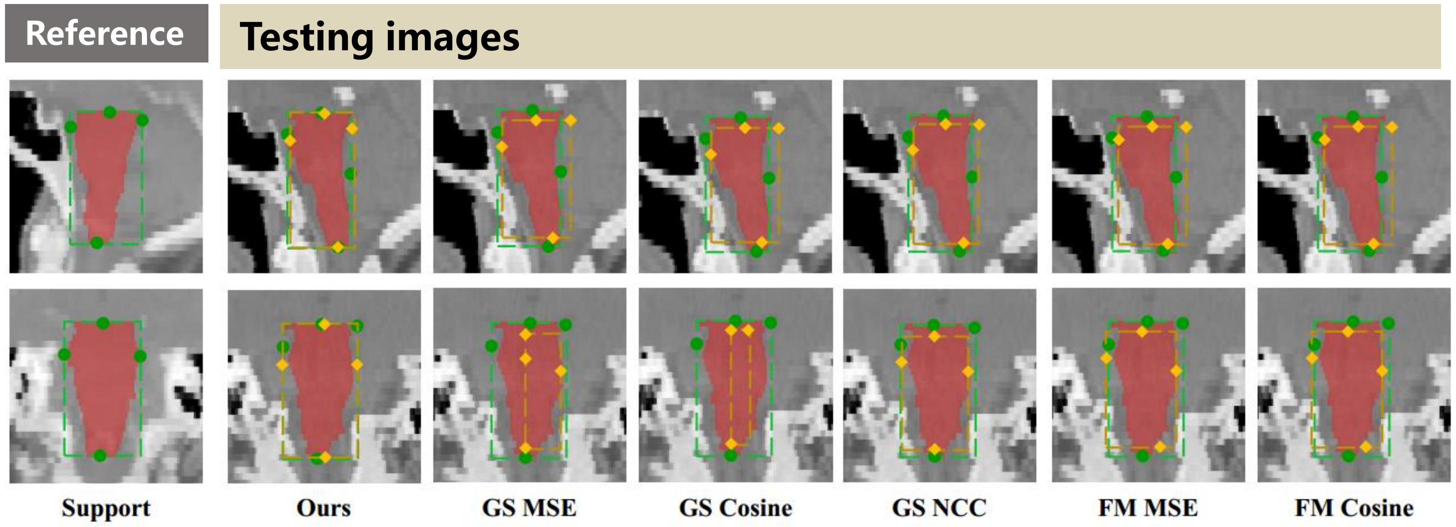
Self-Supervised Learning

- Pretext task: Relative position prediction

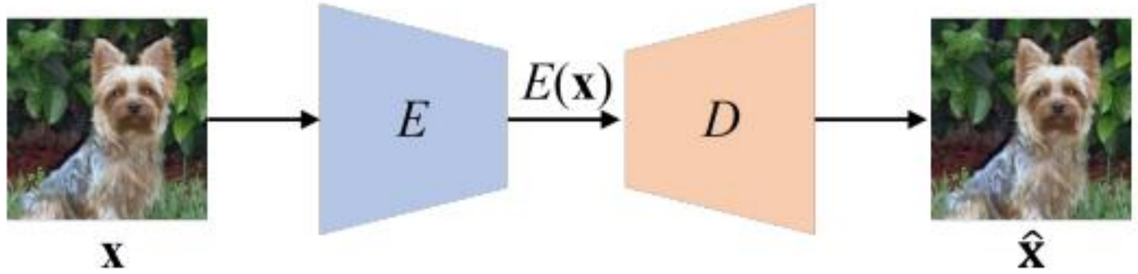
Results on object detection in medical images



Method	Brain stem	Mandible	L Parotid	R Parotid	Mean	Pancreas	Time(s)
Ours	**61.5, 3.70**	**70.0, 5.39**	**44.8, 7.74**	44.2, **7.65**	**55.1, 6.12**	**49.5, 12.22**	**0.15**
GS MSE [1]	39.3, 7.47	65.7, 7.20	39.8, 10.67	37.1, 10.67	45.5, 9.00	1.8, 90.23	1052
GS Cosine [13]	35.2, 7.83	67.4, 7.10	36.9, 11.07	36.7, 10.30	44.1, 9.08	2.7, 118.14	1421
GS NCC [4]	46.9, 7.35	58.9, 9.76	24.3, 18.78	23.9, 20.14	38.5, 14.01	3.5, 101.77	4547
FM MSE [25]	44.7, 6.52	67.7, 6.88	39.2, 10.43	40.6, 8.79	48.1, 8.16	7.9, 120.36	26586
FM Cosine [3]	50.2, 5.91	69.1, 5.53	44.7, 7.97	**44.6**, 8.22	52.2, 7.92	19.2, 56.27	25976
Retina U-Net [8]	68.9, 3.54	75.1, 6.17	60.5, 5.83	63.9, 4.75	67.1, 5.07	81.0, 3.23	4.7

W. Lei et al. Contrastive learning of relative position regression for one-shot object localization in medical images, MICCAI, 2021

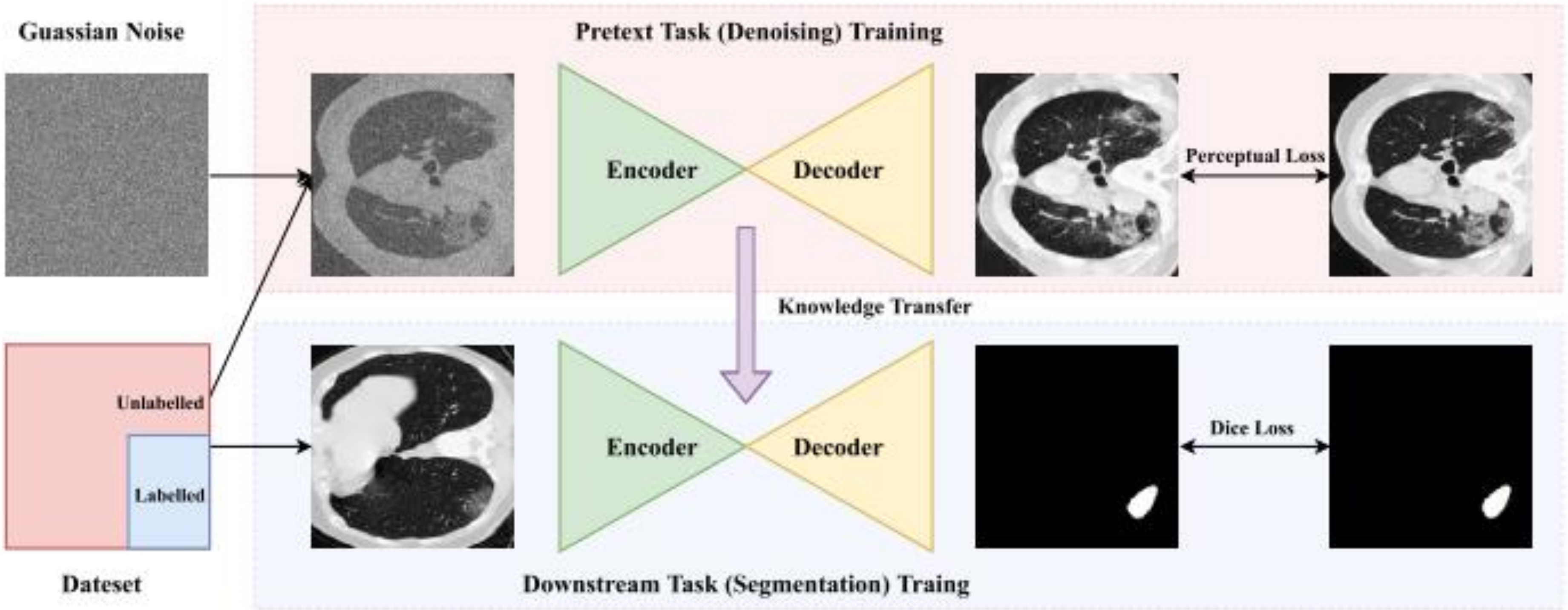
Self-Supervised Learning



- Pretext task: Image Denoising

去噪自编码器

Denoising as a pretext task

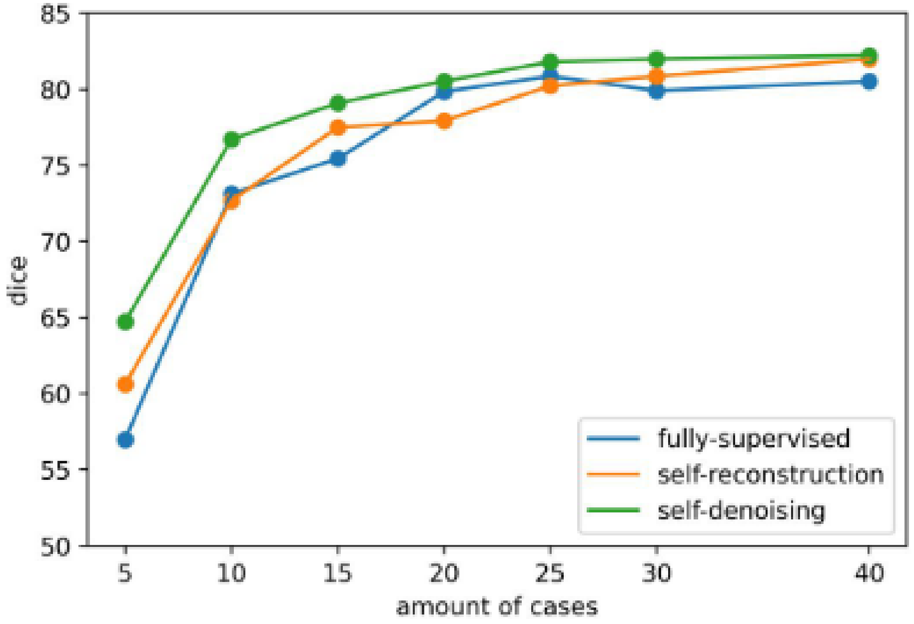


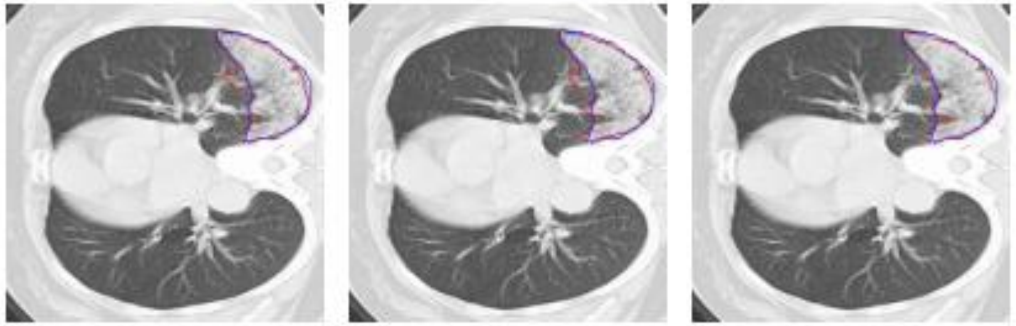
Y. Gao et al. A denoising self-supervised approach for COVID-19 pneumonia lesion segmentation with limited annotated CT images, 2021

Self-Supervised Learning

- Pretext task: Image Denoising

Results





(a) Fully-supervised (b) Self-reconstruction (c) Self-denoising

Method	PSPNet	UNet	UNet++
Fully-Supervised	80.99 ± 10.97	80.52 ± 10.93	82.21 ± 10.21
Self-Denoising	**81.98 ± 10.44**	**82.23 ± 10.03**	**83.35 ± 9.65**

Y. Gao et al. A denoising self-supervised approach for COVID-19 pneumonia lesion segmentation with limited annotated CT images, 2021

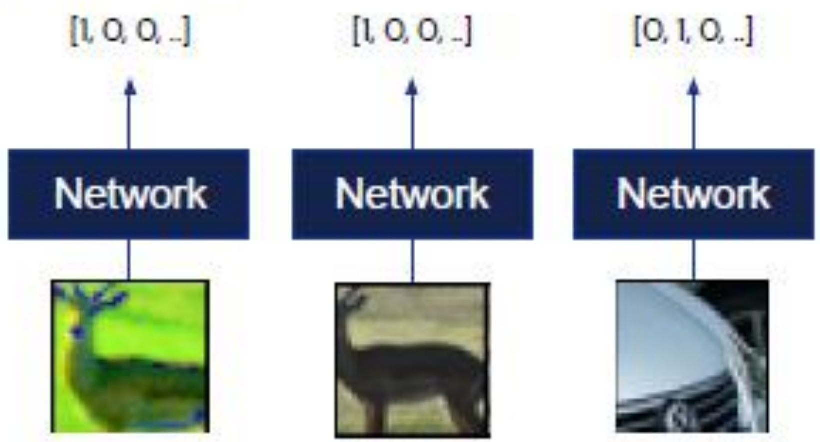
Self-Supervised Learning

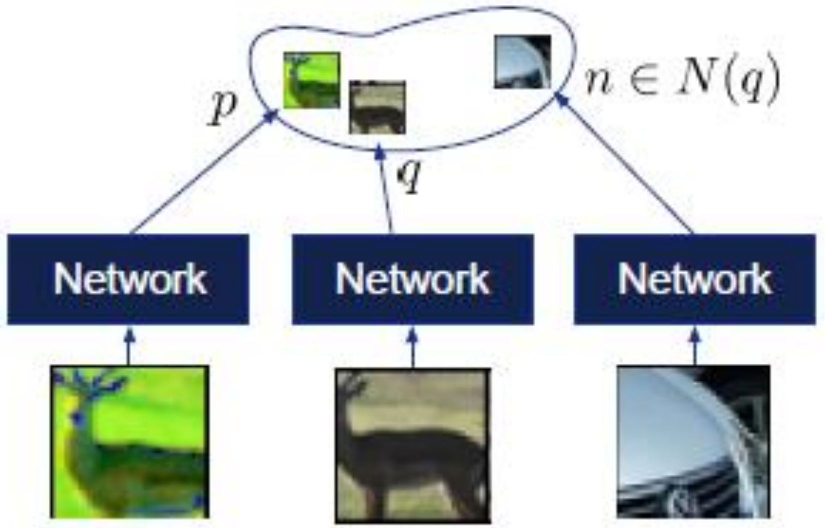
- From Instance classification to metric learning

度量学习

（让模型学会区分不同样本间的相似性）

Classification





只需要训练特征编码器，而无需任务特异性的分类头或者分割/重建解码器

Recently popular: InfoCE

V. Oord et al. Representation learning with contrastive predictive coding, 2018

Self-Supervised Learning

- From Instance classification to metric learning

对比学习

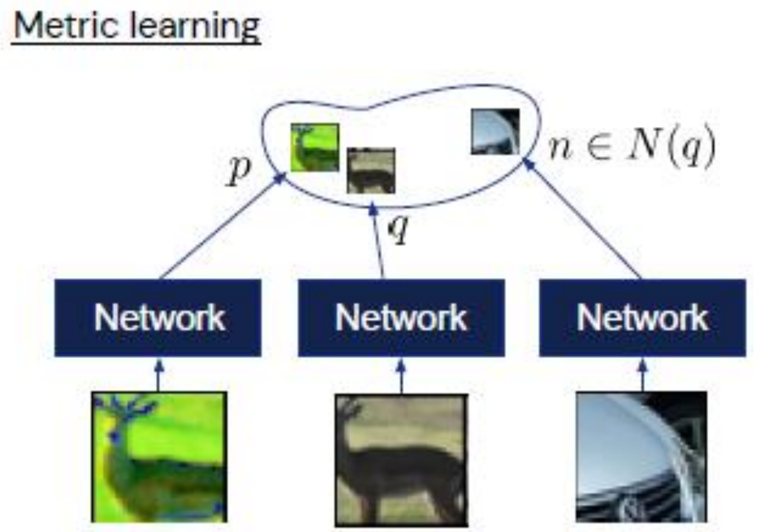
Contrastive Learning: *InfoNCE* loss

- For query, positive and negative:

$$-\log \frac{\exp(q^T p)}{\exp(q^T p) + \sum_{n \in N(q)} \exp(q^T n)}$$

- Link a ranking loss: (q, p) should be close, (q, n) should be far
- An implementation:

$$logits = [q^T p, q^T n_1, q^T n_2, ..] = q^T [p, n_1, n_2, ..]$$
$$labels = [1, 0, 0, ..]$$
$$InfoNCE = cross\_entropy(softmax(logits), labels)$$

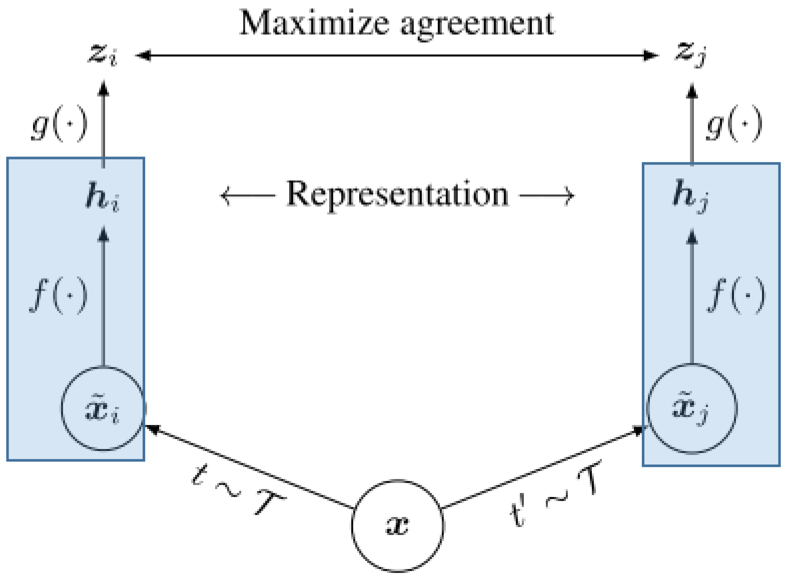


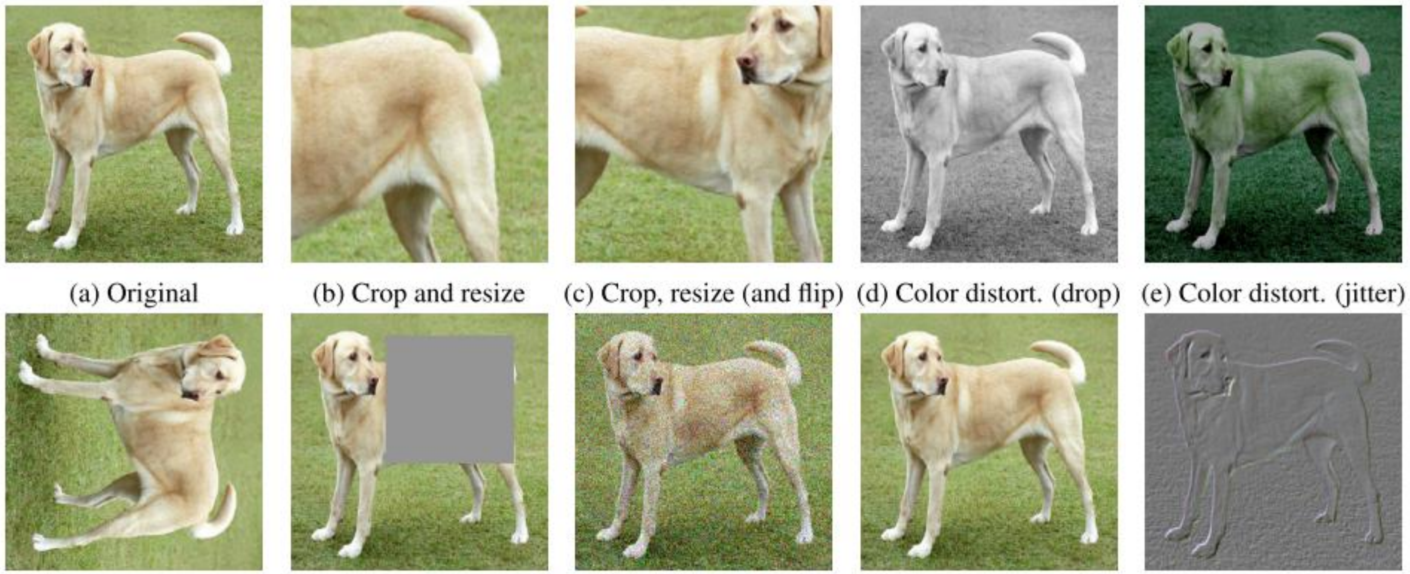
Positive 正样本 | Query | Negative 负样本

V. Oord et al. Representation learning with contrastive predictive coding, 2018

Self-Supervised Learning

- Contrastive learning 对比学习





(a) Original (b) Crop and resize (c) Crop, resize (and flip) (d) Color distort. (drop) (e) Color distort. (jitter)

(f) Rotate $\{90°, 180°, 270°\}$ (g) Cutout (h) Gaussian noise (i) Gaussian blur (j) Sobel filtering

T. Chen et al. A simple framework for contrastive learning of visual representations, 2020

目录

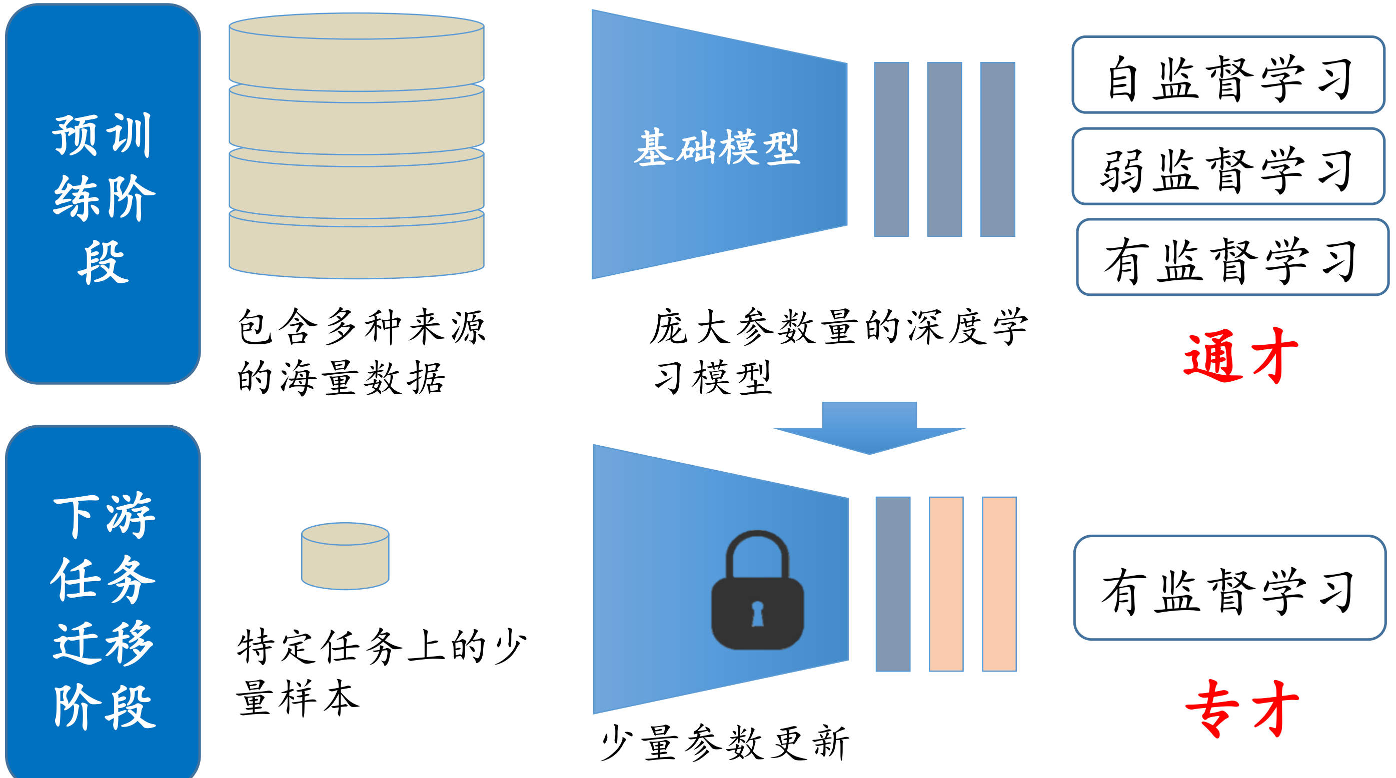
1, Transformer

2，自监督学习

3，视觉基础模型简介

Foundation models

- **Foundation models**: Models that are trained on exorbitant data and compute on a broad task, often intended as a starting point for specialized models



基础模型通过在大规模预训练数据上获取的通用特征：

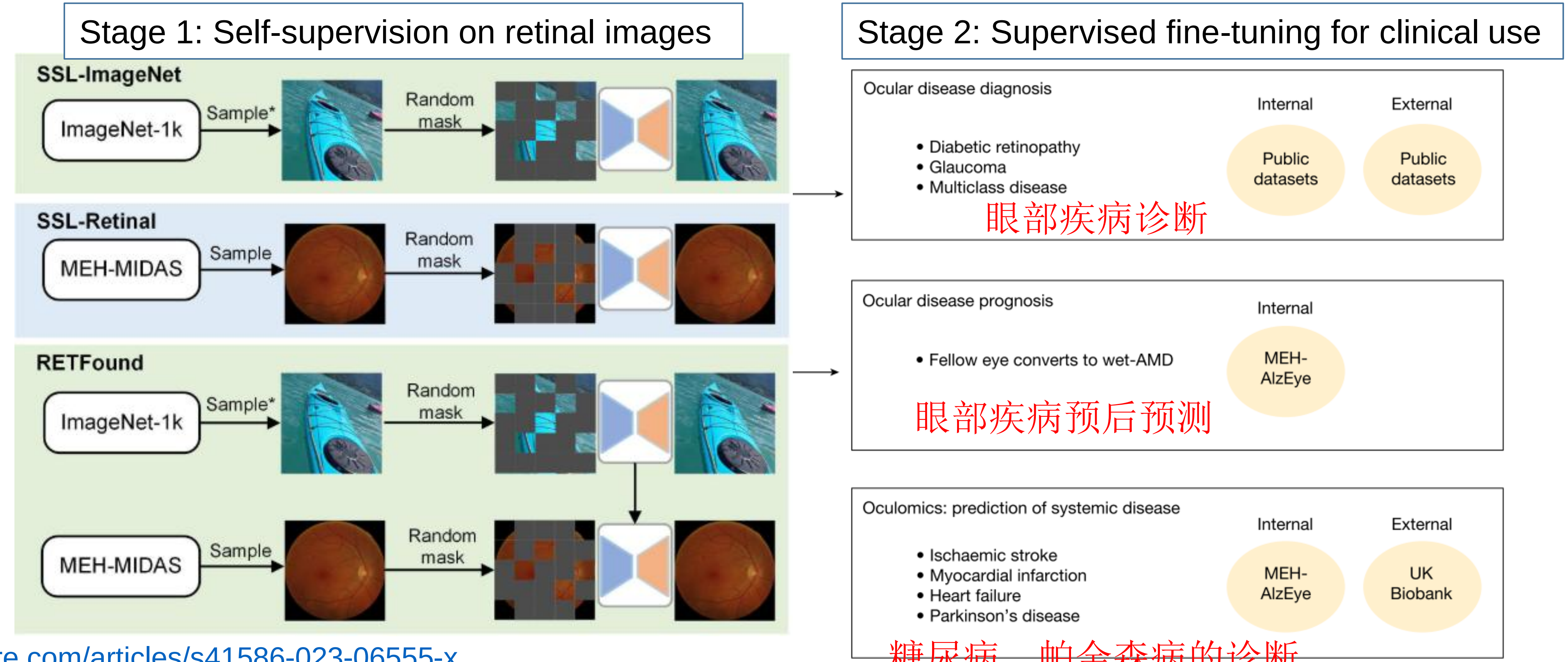
- 有效迁移到下游多种任务中（如图像分类、分割、图文对话等）
- 可以减少下游任务的数据、标注需求，提升下游任务中的性能

Self-Supervised Learning

- Foundation model based on Self-Supervised Learning

在1600万张无标注眼底图像上进行自监督训练（利用MAE）

迁移到多种下游临床诊断任务中



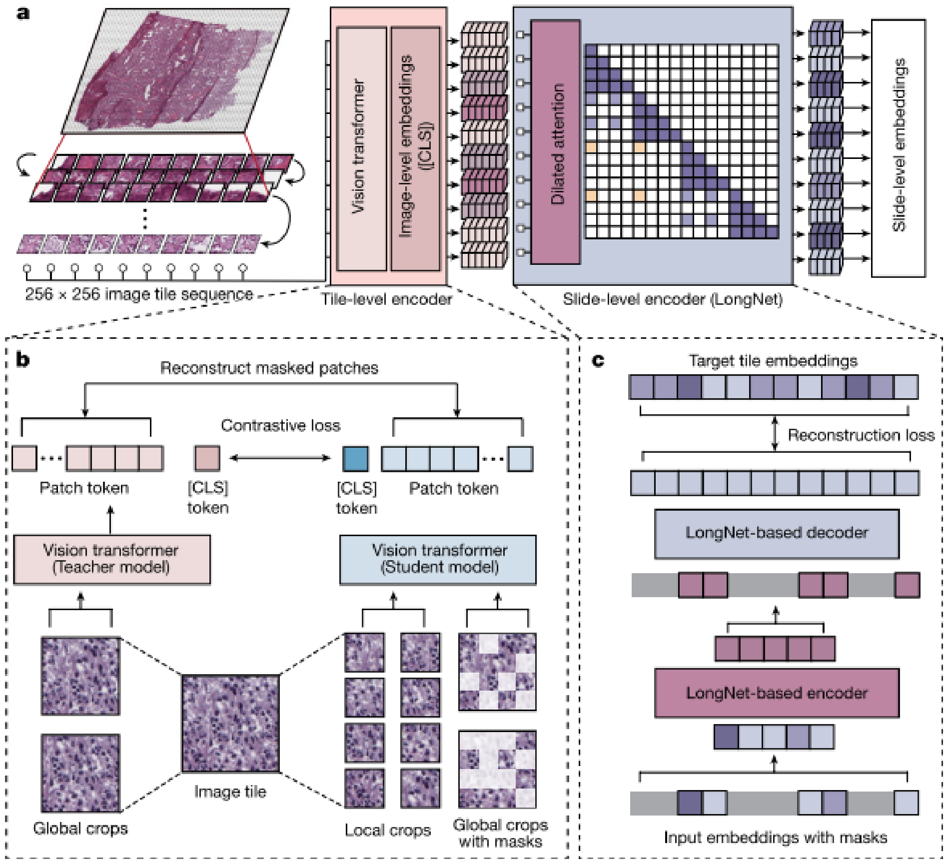
Ariticle: https://www.nature.com/articles/s41586-023-06555-x

Code: https://github.com/openmedlab/RETFound_MAE

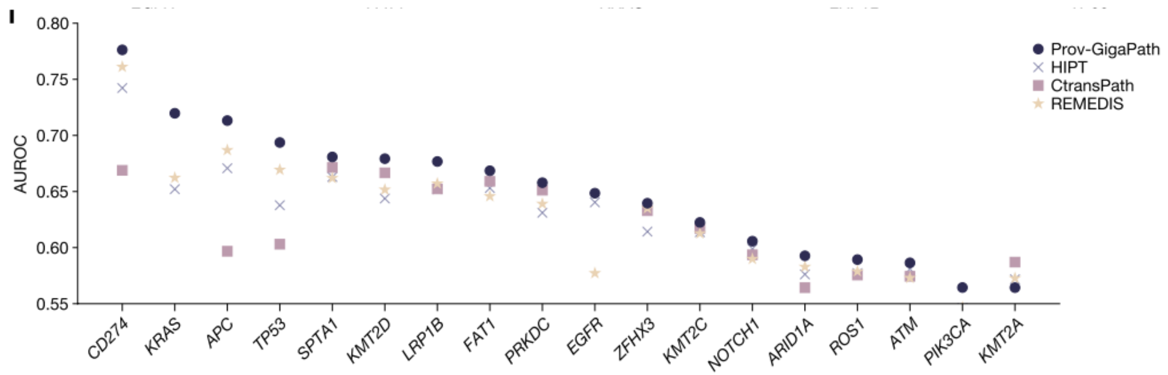
Y. Zhou et al. A foundation model for generalizable disease detection from retinal images, *Nature*, 2023

Self-Supervised Learning

- Digital Pathological Foundation Model (Nature 2024)



1) Training the foundation model with 1.3 billion tiles from 171,189 WSI.
2) DINO-v2 is used for patch-level pretraining,
3) MAE is used for slide-level pretraining



show performance on 9 cancer subtyping tasks and 17 pathomics tasks

A whole-slide foundation model for digital pathology from real-world data, ***Nature***, 2024， Microsoft, Hanwen Xu et al.

Large Multimodal Models (LMM)

多模态模型（视觉语言模型）：结合图片和文字

- 例如：文生图、图生文、图文对话

生成一张秋天的大学校园的照片，有一条大道，两侧是高大的银杏树，满地银杏叶

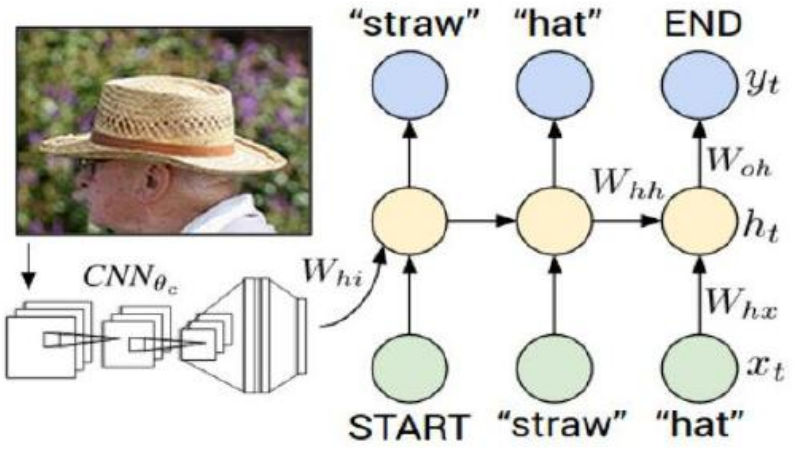
这个图片中搞笑的地方在哪里?

这个图片搞笑的地方在于，它展示了一个极端"不合常理"的场景：一个人站在汽车后备箱上，边开车边熨衣服。正常情况下，熨衣服是需要平稳、安静的环境，而汽车行驶中很颠簸，不适合做这种细致的家务活。此外，这种行为也显得非常危险和不合理，让人觉得好笑。这种反差和荒诞的情境是这张图片的笑点所在。

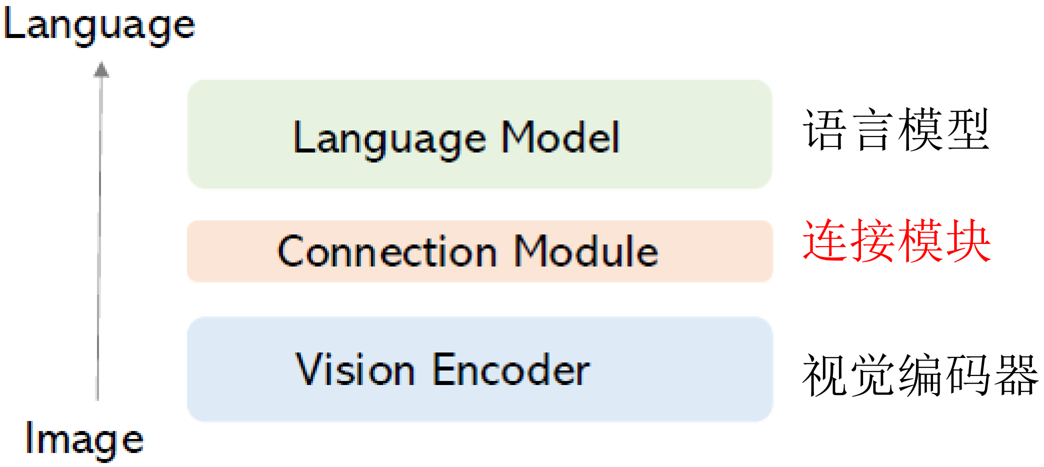
Large Multimodal Models

Model Architectures

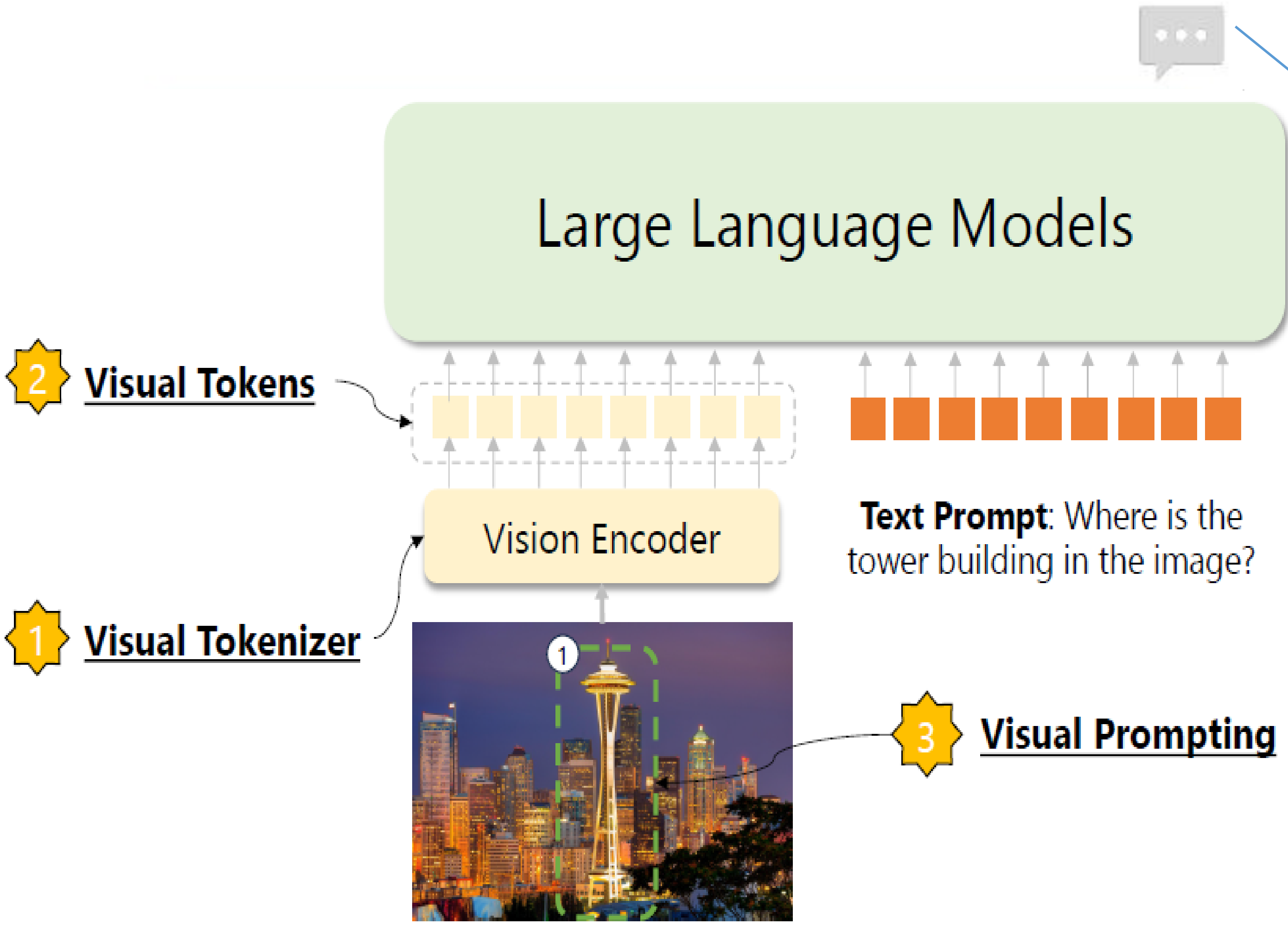
- (Pre-trained) Image Encoder and Language Models
- Trainable modules to connect to two modalities



A dog lying on the grass next to a frisbee

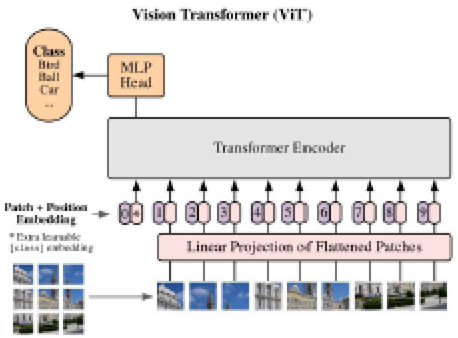


A Typical Large Multimodal Model (LMM)



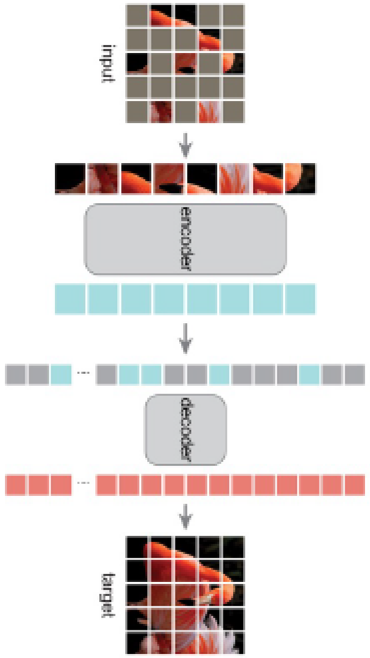
The tower in the image is the **Space Needle**, an iconic observation tower in **Seattle**, Washington, USA. It is one of the **most recognizable landmarks** in the city and offers panoramic views of Seattle and the surrounding area.

An Overview of Vision Encoder



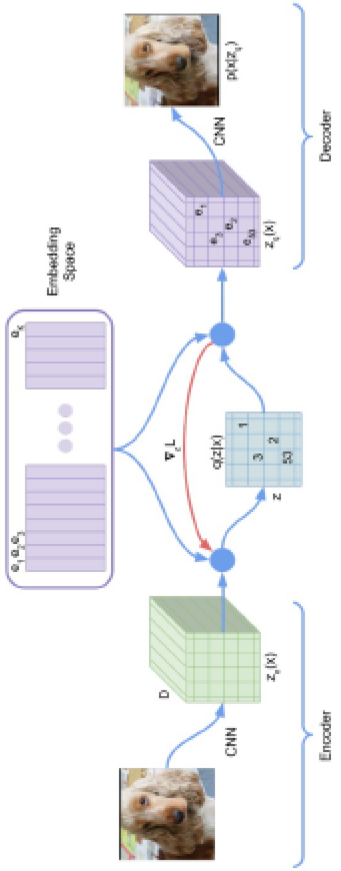
Supervised Learning

ViT, Dosovitskiy et al.



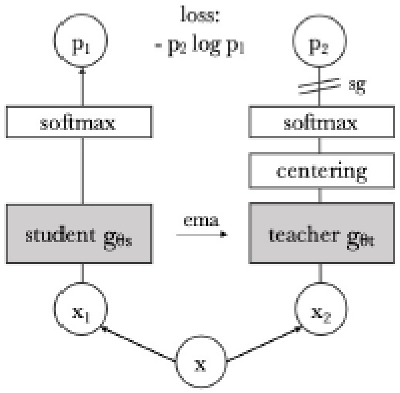
Masked Modeling

MAE, He et al.



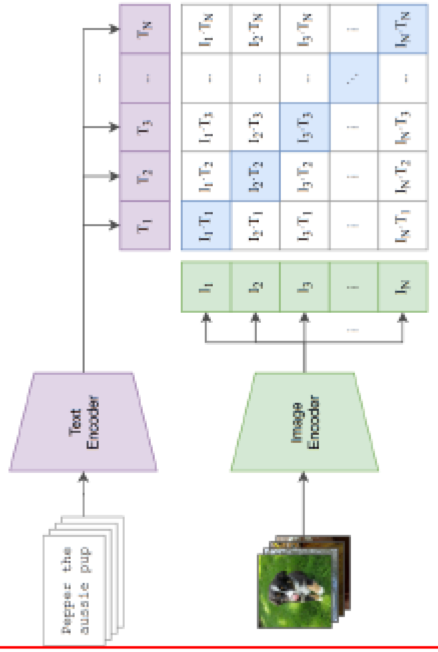
Auto-encoding

VQVAE, Van den Oord et al.



Self-Distillation

DINO, Caron et al.

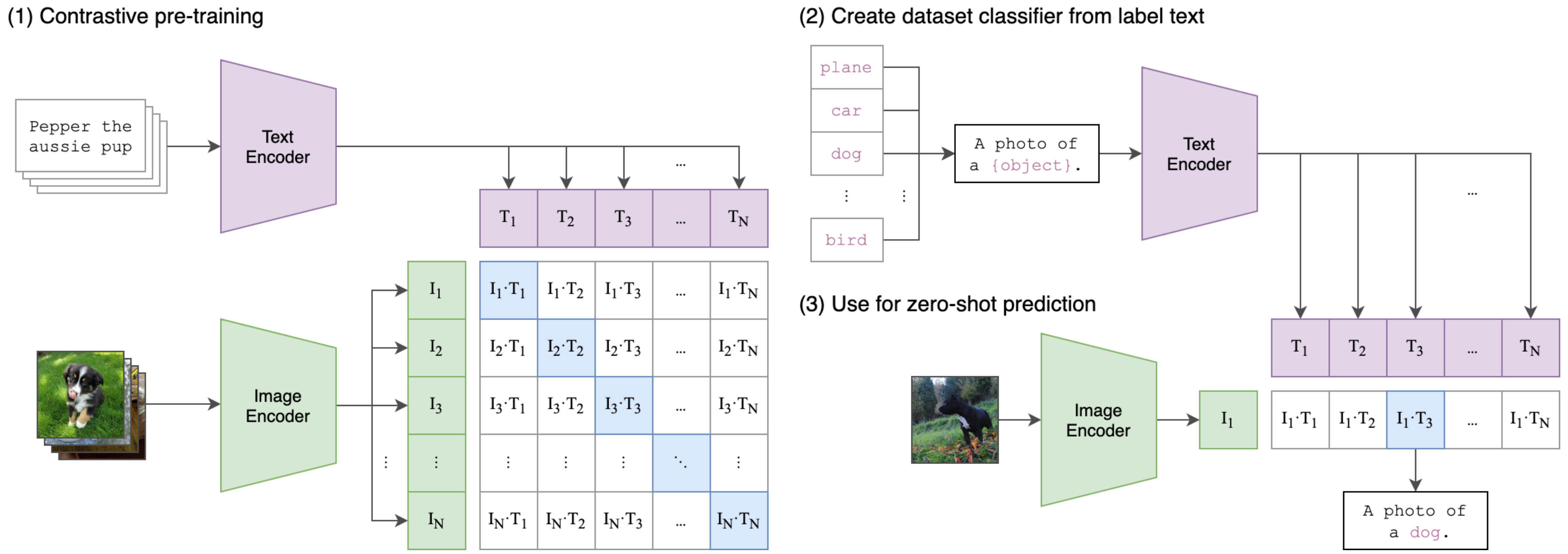


Multimodal Learning

CLIP, Radford et al.

CLIP: Contrastive Language-Image Pre-Training

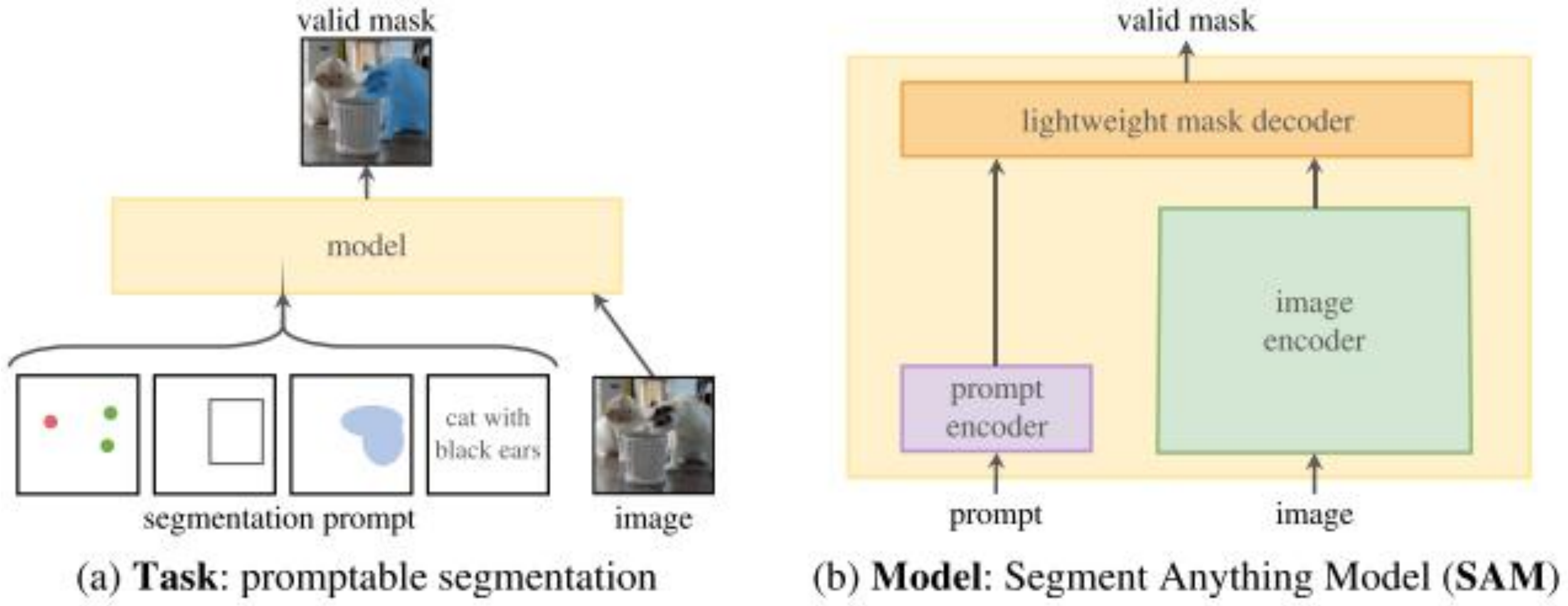
图文对比预训练模型



- Learn from a dataset of 400 million (image, text) pairs
- Strong zero-shot ability

SAM: Segment Anything Model

图像分割基础模型



- Training with over 1 billion masks on 11M images
- Accept multiple kinds of prompts: point, bounding box and mask
- Very strong performance for unseen classes!

Alexander Kirillov et al., Segment Anything, ICCV 2023 (Meta AI research)

SAM: Segment Anything Model

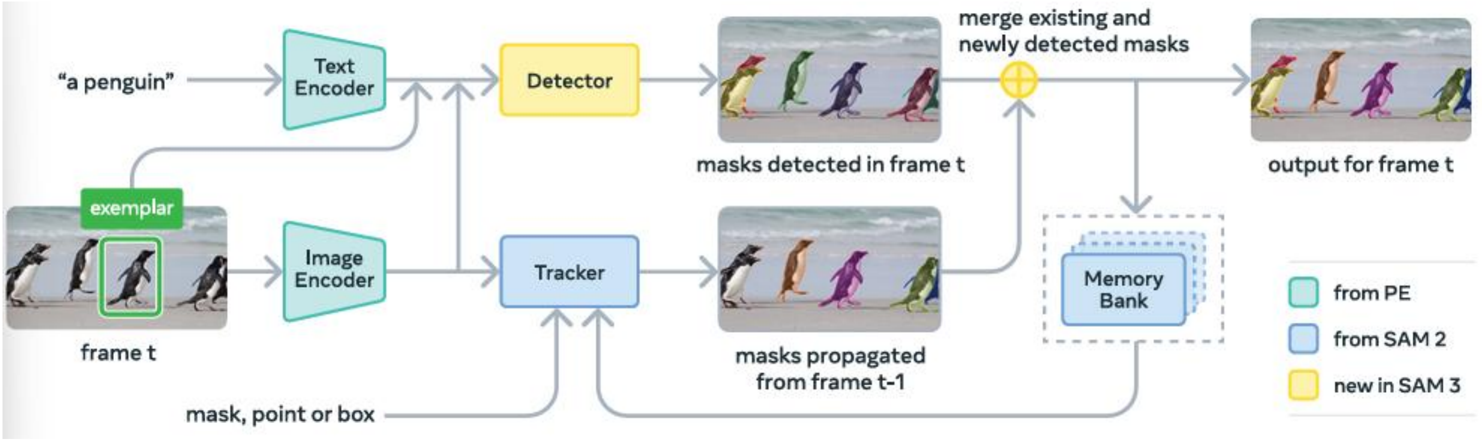
图像分割基础模型

https://segment-anything.com/

SAM 3: Segment Anything with Concepts

基于文本提示，如“yellow school bus”，得到图像中对应物体的分割结果

- 大规模预训练：基于4百万个不重复的概念（concept）标签



Nicolas Carion et al., SAM3: Segment Anything with Concepts, arxiv 2025